CAS Problem – 1 **27-01-2026**

The matrices A and B are given below:

$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Find the eigen values and the associated normalized eigen vectors of the above matrices using R software.

SOLUTION :

```
# create the matrix
A <- c(1,2,2,-2)
B <- c(1,-2,0,-2,5,0,0,0,2)
matA <- matrix(A,nrow = 2, ncol = 2,byrow = TRUE)
matB <- matrix(B,nrow = 3,ncol = 3,byrow= TRUE)

# Gives the eigen value and eigen vector
eigen(matA)

## eigen() decomposition
## $values
## [1]  2 -3
##
## $vectors
##            [,1]       [,2]
## [1,] -0.8944272 -0.4472136
## [2,] -0.4472136  0.8944272

eigen(matB)

## eigen() decomposition
## $values
## [1] 5.8284271 2.0000000 0.1715729
##
## $vectors
##            [,1] [,2]      [,3]
## [1,] -0.3826834    0 0.9238795
## [2,]  0.9238795    0 0.3826834
## [3,]  0.0000000    1 0.0000000
```